

NYU Langone

How might we uplift the healthcare journey for patients with limited English proficiency (LEP) in New York City, addressing language barriers and cultural nuances to ensure equitable access to healthcare services?



Problem Statement

Navigating the fragmented U.S. healthcare system is challenging for patients with limited English proficiency (LEP).

Large Population

36%

Approximately 36.3% of the population in New York City is foreign-born.

Source

Miscommunication

50%

Approximately 50% of the medical information is remembered incorrectly by patients.

Source

Lack of Support Service

13%

Only 13% of hospitals are compliant with National Standards for Culturally and Linguistically Appropriate Services (CLAS) in health care.

Source

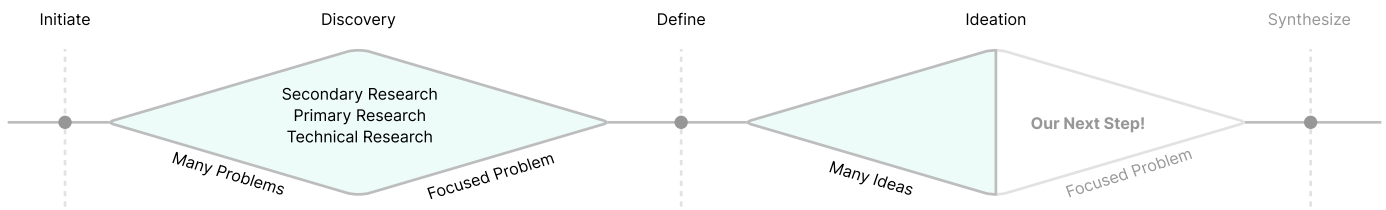
Long waiting time

100_{min}

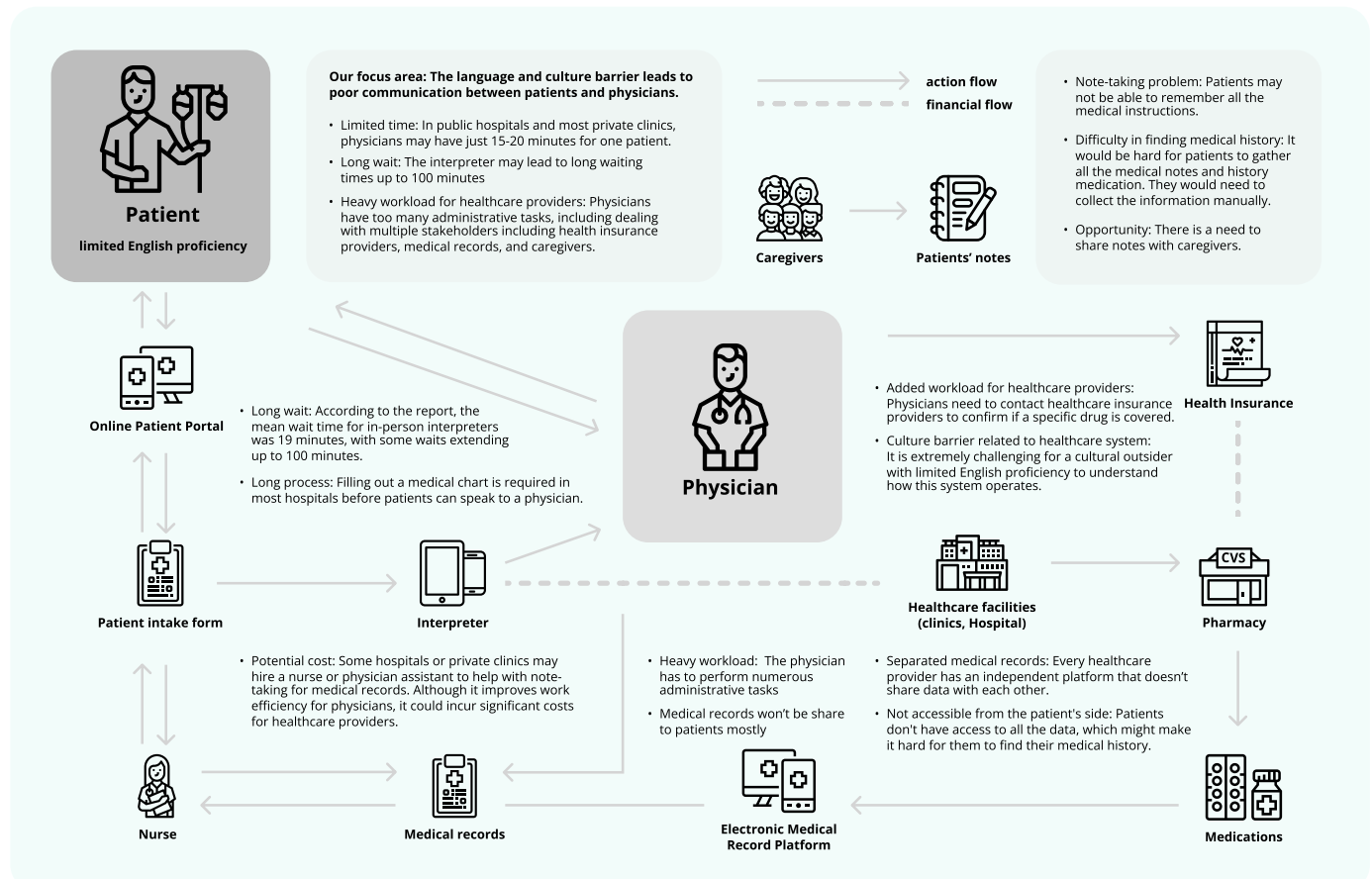
the mean wait time for in-person interpreters was 19 minutes, up to 100 minutes.

Source

Research Process



System Map



Secondary Research

We looked into reports, research papers, existing and emerging products, to gain a holistic understanding of the challenges healthcare system faces and identify emerging opportunities.

Understanding the U.S. Healthcare System

As stated in the design brief and other secondary sources we collected, the U.S. healthcare system faces many systemic challenges, involving multiple stakeholders and complex procedures.

How Exactly Emerging technology Like AI Can Help

Digital healthcare is a significant industry trend, with AI playing a transformative role. We explored AI capabilities and healthcare & AI applications and startups within the healthcare sector.

[More on Secondary Research](#)

Challenges for Patients with limited English proficiency

Patients with LEP in the United States face more difficulties navigating the healthcare system due to language barriers, cultural differences, unfamiliarity with the system, economic barriers, and eligibility issues.

Product Positioning and Business Viability

We conducted an analysis of potential AI use cases and evaluated current and emerging products to understand their business models, enabling us to better position our design solution.

Competitive Analysis

We began with a general competitive analysis of existing and emerging solutions, then reframed the analysis based on the following aspects:

- Patient Centric (User Experience)
- Clinical Impact
- Technical Performance & Legal Compliance
- Integration with Healthcare System
- Innovation and Scalability

Company	Product	Key Features	Target Audience	Market Position	Competitive Advantage	Challenges	Opportunities	Future Outlook
Google	Google Assistant	Voice search, smart home integration	General consumer	Market leader	Large ecosystem, AI capabilities	Privacy concerns, limited healthcare integration	Healthcare applications, improved accuracy	Continued growth in AI and voice search
Amazon	Alexa	Voice assistant, smart home control	General consumer	Strong competitor	Wide device availability, ecosystem	Similar to Google Assistant	Healthcare-focused skills, improved voice recognition	Expanding into healthcare and professional services
Apple	Siri	Voice assistant, integrated with iOS	iOS users	Established player	Deep integration with Apple ecosystem	Limited third-party integration	Healthcare integration, improved Siri capabilities	Continued innovation in AI and ecosystem integration
Microsoft	Cortana	Voice assistant, productivity tools	Windows and Office users	Established player	Integration with Windows and Office	Limited third-party integration	Healthcare integration, improved productivity features	Continued focus on productivity and AI integration
IBM	Watson	AI-powered healthcare solutions	Healthcare providers	Enterprise focus	Advanced AI capabilities, healthcare expertise	High cost, complex integration	Improved accuracy, broader healthcare applications	Continued leadership in enterprise AI
Oracle	HealthInsight	AI-powered analytics for healthcare	Healthcare providers	Enterprise focus	Strong data capabilities, healthcare expertise	High cost, complex integration	Improved analytics, broader healthcare applications	Continued focus on enterprise AI and healthcare
Microsoft	Healthcare AI	AI-powered healthcare solutions	Healthcare providers	Enterprise focus	Advanced AI capabilities, healthcare expertise	High cost, complex integration	Improved accuracy, broader healthcare applications	Continued leadership in enterprise AI
Google	Healthcare AI	AI-powered healthcare solutions	Healthcare providers	Enterprise focus	Advanced AI capabilities, healthcare expertise	High cost, complex integration	Improved accuracy, broader healthcare applications	Continued leadership in enterprise AI
Amazon	Healthcare AI	AI-powered healthcare solutions	Healthcare providers	Enterprise focus	Advanced AI capabilities, healthcare expertise	High cost, complex integration	Improved accuracy, broader healthcare applications	Continued leadership in enterprise AI
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	Cognigy AI	Authenticx	Mediktor	Zocdoc	Bamboo health	Busy Health	Smartact ion	Aisera	Welldoc	Dragon Ambient eXperience	Augmedix	Suki
Patient Centric (User Experience)	enhanced experiences across various communication channels.	Agent Performance Dashboard	Detailed Pre-Diagnosis Guidance Multi language support	Streamlined Doctor Search, compare appointment	Care Collaboration Platform Patient Screening Solution Healthcare Technology Solutions	Detailed Pre-Diagnosis Guidance Community Building Clear and Informative Communication	personalized recommendation	personalized recommendation	Multiple device integration	N/A	N/A	N/A
Clinical Impact	N/A	N/A	Improve pre-diagnosis accuracy	Improve medical reach	Automated User Management	Extensive Symptom Inquiry	N/A	N/A	N/A	Enhance clinical efficiency	help patient prepare for doctor visit Enhance clinical efficiency	Enhance clinical efficiency
Technical Performance & Legal Compliance	Multiple platform Low-Code Editor Live Transcript	Information Security & Compliance Quality Assurance Solution Large scale analysis	Apple Health Integration Clinical Validated	N/A	N/A	N/A	HIPAA compliant	HIPAA compliant	FDA-cleared and designed for managing chronic diseases	Multiple platform Data security protocols	HIPAA compliant	Multiple platform SOC2 Type 2 certified HIPAA compliant all time available 24/7/365
Integration with Healthcare System	N/A	N/A	Integration with multiple stakeholders: insurers, hospital, pharmacy.	partnership with insurance & medical provider EHR integration	Extensive Network	N/A	billing management make payments appointment scheduling EHR integration	N/A	N/A	Integration with 200+ EHR systems	Integration with 50+ EHR systems	Integration with major EHR systems
Innovation and Scalability	advanced contextual processing abilities Emotion detection Pretrained Skills	N/A	AI for pre-diagnosis & Triage	N/A	N/A	Interactive Chat-Driven Platform	conversational AI based on NLP	NLP conversational AI	Realtime and actionable insights for patient	conversational AI real-time voice translation ambient medical documentation NLP	chart preparation before patient visit conversational AI real-time voice translation ambient medical documentation NLP	conversational AI real-time voice translation ambient medical documentation NLP

[More on Competitors Research](#)

Connected to the Primary Research

As we conducted interviews, we incorporated the selected product features from research and found their desirability in real-world scenarios.

Connected to the Technical Research

Our two developers on the team conducted further research on these features and their usage, helping us understand their technical feasibility.

Primary Research

We conducted 20 interviews to gain an in-depth understanding of the pain points people may experience during the process of seeing a healthcare provider.

16 Non-native speakers who have lived in New York and have had medical experiences within the US

4 Healthcare providers experienced in treating patients with limited English proficiency (LEP)

Interview Objectives

To identify communication and navigation challenges faced by non-native speakers and healthcare professionals within the healthcare system.

Categories of Questions

- Overall Healthcare Visit Experience
- Communication Challenges
- Note-Taking
- Knowledge and Understanding of Insurance
- Product Idea Feedback

Analysis Process

Affinity Mapping

We utilized affinity mapping to categorize the insights from our interviews.

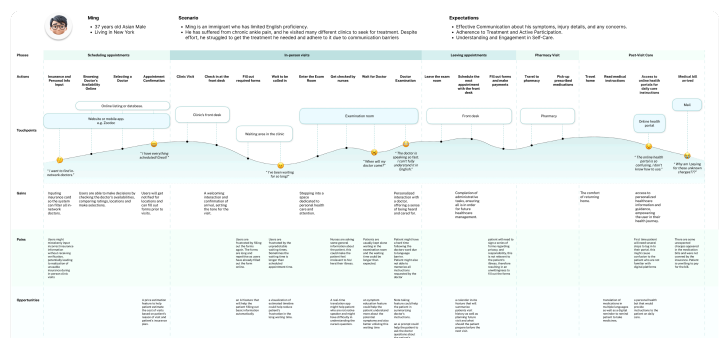
[More about Affinity Map](#)



Journey Map

We created a user journey map to identify potential opportunities that will guide our strategic direction.

[More about Journey Map](#)



Key Takeaways

Communication and Cultural Barriers

Language barriers lead to communication delays and misunderstandings. Even though translation tools are available, there is a strong preference for direct communication.

Desirability for a Digital Medical Assistant Tool

There is strong interest in digital tools for improving communication and treatment efficiency, with desired features such as translation, note-taking, and patient education.

Difficulty in Memorizing Doctor's Instructions

Patients utilize both digital and handwritten notes to recall medical advice but frequently find themselves needing to research terms after the visit for clarity.

Unfamiliarity with the Healthcare System

Patients demonstrate a limited understanding of insurance details, relying on healthcare providers for information and prioritizing treatment over navigating insurance complexities.

Technical Research

Through primary (user pain points) and secondary research, we have pinpointed features that necessitate an evaluation of their technical feasibility.

Language Identification (LID)

LID, or Language Identification, refers to the process of automatically identifying the language of a given text or speech.

- [fasttext LID](#)
- [lingua](#)
- [cld3](#)
- [tika](#)

Potential Uses in Our Project

- Accurately determine the language for non-native patients/ limited english proficiency patients.

Real-time Transcription

Real-time transcription refers to the immediate and simultaneous conversion of spoken language into written text.

- [DeepSpeech](#)
- [kaldi](#)
- [pocketsphinx](#)
- [vosk-api](#)
- [wav2vec2](#)

Potential Uses in Our Project

- Transcribing prescriptions and conversations between patients and healthcare providers.

Real-time Translation

Real-time translation is a technological advancement that enables the immediate conversion of spoken text from one language to another with low delay.

- [fairseq](#)
- [argos-translate](#)
- [marian](#)
- [bergamot-translator](#)

Potential Uses in Our Project

- Facilitating the communication between patients and doctors.

Large Language Models (LLMs)

LLMs have significant potential due to their ability to process and understand vast amounts of text-based information. They can be trained on extensive medical literature, including textbooks, research papers, and clinical guidelines.

- [biobert](#)
- [UmlsBERT](#) - Unified Medical Language System
- [clinicalBERT](#) - Clinical Notes
- [bluebert](#)

Potential Uses in Our Project

- Analyzing the symptoms described by patients.
- Give suggestions based on the symptoms.
- Advise on the urgency of seeking medical attention.

Recommendation System

Recommendation system in medical context leverages data analysis including medical history, current conditions and uses machine learning to provide personalized healthcare service recommendations.

- [lightfm](#)
- [Surprise](#)
- [tensorflow-recommenders](#)

Potential Uses in Our Project

- Providing patient with medical suggestion, learning content, etc.

Time Series Forecasting

Time series forecasting is a statistical technique for predicting future events and stats by analyzing the patterns in historical data.

- [prophet](#)
- [pytorch-forecasting](#)
- [gluon-ts](#)
- [statsmodels](#) (ARIMA, SARIMA)

Potential Uses in Our Project

- Bill price prediction.
- Chronic condition monitoring.

Our next step

Moving forward, we plan to ideate on possible solutions and conduct a second round of quantitative & qualitative research.

- This next phase will include a broader demographic of patients with limited English proficiency who have experienced New York's fragmented healthcare services.